

Assignment 2 — E-Commerce Sales Analysis

Dataset

ecommerce_orders_raw.csv

Objective

You are a data analyst for an e-commerce company. The company wants to understand its customers, products, sales performance, and revenue trends. Use Pandas to transform the raw order data into meaningful business information.

Part A — Data Understanding

1. Load the dataset.
2. Display its structure and dimensions.
3. Identify the data types of all columns.
4. Identify missing values.
5. Identify duplicate orders.
6. Examine the unique values in:
 - City
 - Product Category
 - Payment Method

Part B — Data Cleaning

7. Handle missing ratings.
8. Standardize city names.
9. Identify invalid quantities.
10. Handle duplicate orders.
11. Convert Order_Date into an appropriate datetime format.

Part C — Feature Engineering

12. Create:

$\text{Gross_Amount} = \text{Quantity} \times \text{Unit_Price}$

13. Create:

$\text{Discount_Amount} = \text{Gross_Amount} \times \text{Discount} / 100$

14. Create:

$\text{Net_Amount} = \text{Gross_Amount} - \text{Discount_Amount}$

15. Create a Rating_Category column:

1–2 → Poor
3 → Average

4 → Good
5 → Excellent

16. Extract the following from Order_Date:

- Year
- Month
- Day
- Day of Week

Part D — Business Analysis

17. Calculate total revenue.
18. Calculate total number of orders.
19. Find the average order value.
20. Find total revenue by product category.
21. Find the top 10 products by revenue.
22. Find the city generating the highest revenue.
23. Find the most frequently used payment method.
24. Find the average rating for each product category.
25. Calculate monthly revenue.
26. Identify the highest-revenue month.
27. Find the top 10 customers by total spending.
28. Compare revenue generated by different cities.
29. Find which product category has the highest average order value.